NEXUS: A Hybrid Employability Evaluation Framework via Deterministic Matching, NLP Semantics, and Prerequisite Directed Acyclic Graphs

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Abstract

This paper presents NEXUS (Next-generation Employability, eXpertise & Upskilling System), a multi-tier career readiness framework that integrates (i) deterministic proficiency-weighted skill matching with overqualification-capped scoring, (ii) sentence-transformer NLP semantic cosine verification, (iii) a novel 2-pass LLM disagreement reconciliation loop with dynamic weight shifting, (iv) prerequisite-aware topological learning path generation via directed acyclic graphs, and (v) a heuristic time-to-readiness estimator with continuous closed-loop feedback. Experimental verification on a live student-job matching instance demonstrates 67% weighted match, 74.2% semantic score, and reconciliation convergence within 2 LLM passes. The framework addresses fundamental limitations in existing keyword-based ATS platforms through mathematically grounded, multi-signal fusion architecture.

Keywords: *Weighted Skill Matching, Sentence Transformers, LLM Reconciliation, DAG, Cosine Similarity, Career Readiness*

I. INTRODUCTION

Existing Applicant Tracking Systems (ATS) rely on boolean keyword matching, treating skills as binary attributes (present/absent). This approach fails to capture proficiency depth, skill importance hierarchies, and contextual relevance. NEXUS addresses these gaps through a mathematically grounded multi-signal fusion architecture combining deterministic scoring, NLP semantics, and LLM-based evidence evaluation with a novel disagreement reconciliation mechanism.

The key contributions of this work are: (1) a weighted proficiency matcher with overqualification capping, (2) dual-layer NLP with transformer-to-TF-IDF fallback, (3) a 2-pass LLM reconciliation loop with dynamic weight shifting, (4) prerequisite DAG-based topological learning paths, and (5) a continuous closed-loop skill elevation feedback system.

II. SYSTEM ARCHITECTURE

NEXUS operates across three computational layers connected by a closed-loop feedback channel, as depicted in Fig. 1.

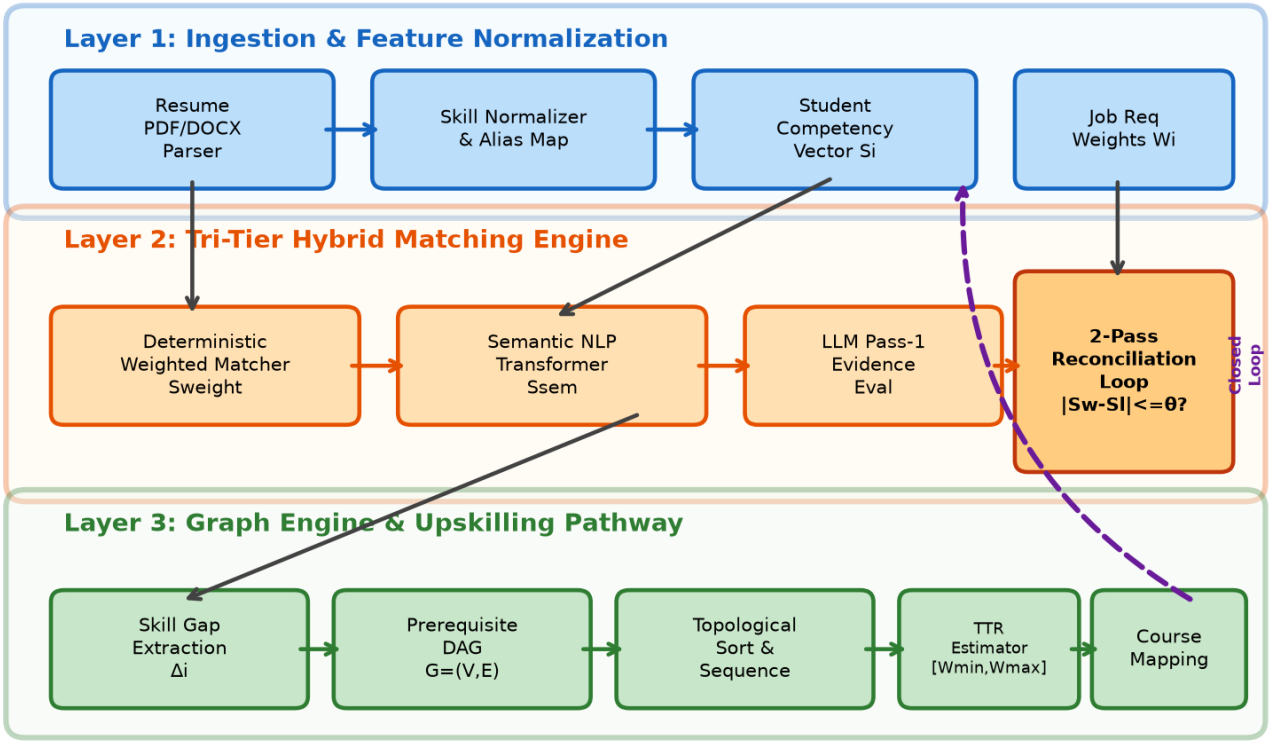


Fig. 1: NEXUS Three-Layer Architecture with Closed-Loop Feedback

III. MATHEMATICAL FORMULATIONS

A. Deterministic Weighted Skill Matcher

For a job requiring n skills, each with required proficiency Ri, importance weight wi, and student proficiency Si, the individual skill score is capped to prevent overqualification inflation:

$$\sigma_i = \min\left(\frac{S_i}{R_i},\ 1.0\right)$$
$$S_{\text{weight}} = \sum_{i=1}^{n} \sigma_i \cdot w_i \cdot 100$$
$$\text{subject to} \sum_{i=1}^{n} w_i = 1.0$$

Eq. 1: Weighted Match Score with Overqualification Cap

The skill gap and mandatory violation indicator are defined as:

$$\Delta_i = \max(R_i - S_i,\ 0)$$
$$\Phi_{\text{mand}} = \bigvee_{i=1}^{n} [\mathbb{I}(m_i = 1) \wedge (\Delta_i > 0)]$$

Eq. 2: Gap Function & Mandatory Skill Violation Flag

B. Semantic NLP Embedding & Cosine Verification

Resume and job description texts are projected into 384-dimensional dense space using the all-MiniLM-L6-v2 sentence transformer. Cosine similarity measures contextual alignment:

$$\mathbf{u} = \text{Encoder}(\mathbf{T}_{\text{resume}}),\quad \mathbf{v} = \text{Encoder}(\mathbf{T}_{\text{job}})$$

$$\text{Sim}_{\cos}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|_2 \cdot \|\mathbf{v}\|_2}$$

$$S_{\text{sem}} = \text{clamp}(\text{Sim}_{\cos} \times 100,\ 0,\ 100)$$

Eq. 3: Sentence Transformer Embedding & Cosine Similarity Score

C. Two-Pass LLM Reconciliation Loop

The hybrid evaluator cross-validates deterministic and LLM scores. If disagreement exceeds threshold theta, a second strict pass triggers with dynamic weight rebalancing toward the deterministic score (trust-but-verify principle):

$$\text{AGREED}: \ S_f = 0.50\, S_w + 0.25\, S_s + 0.25\, S_\ell^{(1)}$$

$$\text{RECONCILED}: \ S_f = 0.60\, S_w + 0.20\, S_s + 0.20\, S_\ell^{(2)}$$

$$\text{REVIEW_REQ}: \ S_f = 0.70\, S_w + 0.15\, S_s + 0.15\, S_\ell^{(2)}$$

Eq. 4: Dynamic Weight Composition by Validation State

Fig. 2: Two-Pass LLM Reconciliation Loop with Dynamic Weight Shifting

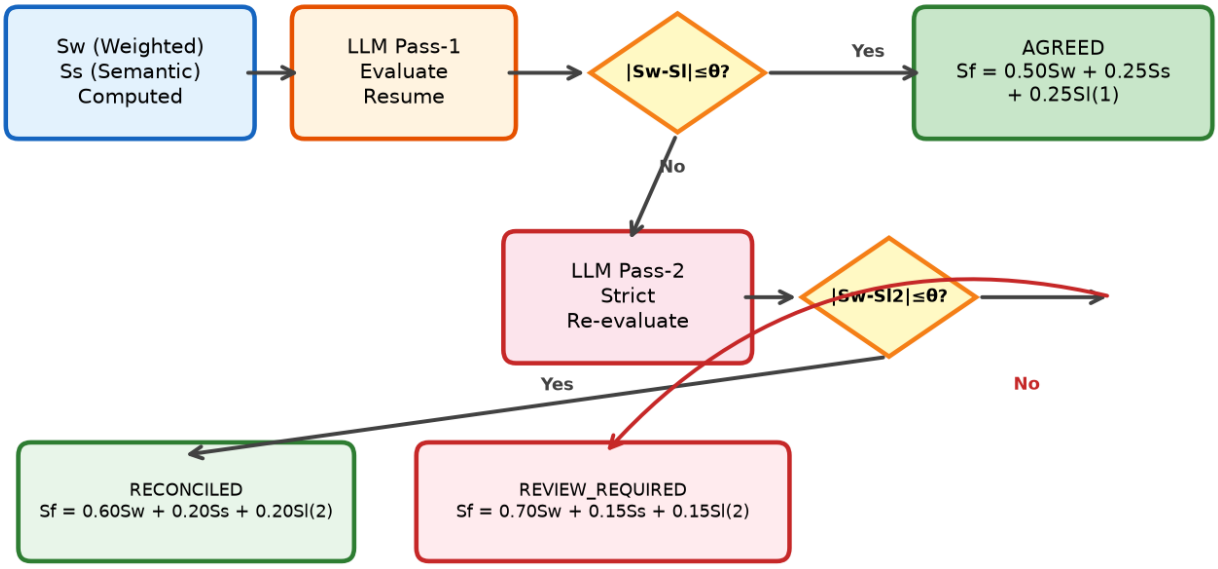


Fig. 2: Two-Pass LLM Reconciliation State Machine

D. Prerequisite DAG & Topological Ordering

Skills are modeled as a DAG G = (V, E) where edge (u, v) indicates u must be learned before v. Cycle detection validates DAG integrity; topological sort generates dependency-respecting sequences:

$$\text{Cycles}(\mathcal{G}) = \emptyset \implies \exists\, \pi : \mathcal{V} \to \{1, \ldots, |\mathcal{V}|\}$$

$$\text{s.t.} \quad (u, v) \in \mathcal{E} \Rightarrow \pi(u) < \pi(v)$$

Eq. 5: Topological Ordering Existence Condition

Fig. 3: Skill Prerequisite DAG G = (V, E) with Status Coloring

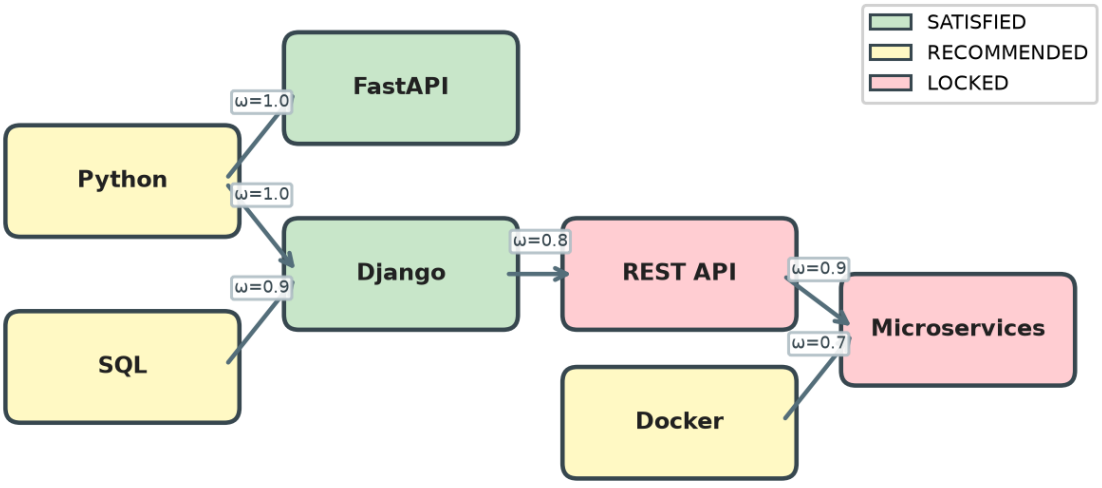


Fig. 3: Skill Prerequisite DAG with Node Status Coloring

E. Time-to-Readiness (TTR) Heuristic Estimator

TTR estimates realistic week ranges based on gap magnitude, skill difficulty, prerequisite depth complexity, and allocated weekly study hours:

$$H_{\text{base}} = \sum_{j} \Delta_j \cdot (10 + 2D_j)$$

$$H_{\text{total}} = H_{\text{base}} \cdot (1 + 0.10 \cdot \delta_{\text{DAG}})$$

$$W_{\min} = \max\left(1,\ \lfloor 0.85 \cdot H_{\text{total}} / H_w \rfloor\right)$$

$$W_{\max} = \max\left(W_{\min} + 1,\ \lceil 1.25 \cdot H_{\text{total}} / H_w \rceil\right)$$

Eq. 6: Time-to-Readiness Range Estimation Model

F. Job Match Probability Predictor

Final match probability integrates weighted score, semantic score, mandatory satisfaction, and CGPA eligibility with multiplicative penalty factors:

$$P_{\text{base}} = 0.70 \cdot S_w + 0.30 \cdot S_s$$

$$P_{\text{match}} = P_{\text{base}} \cdot 0.50^{\mathbb{I}(\Phi_{\text{mand}})} \cdot 0.80^{\mathbb{I}(\text{CGPA} < \text{CGPA}_{\min})}$$

Eq. 7: Match Probability with Penalty Factors

IV. RELATIONAL DATABASE SCHEMA (3NF)

NEXUS uses a 14-table 3NF-normalized SQLite schema. Key entities include STUDENT, SKILL (with SKILL_ALIAS for NLP normalization), JOB with weighted JOB_SKILL requirements, COURSE mapped to skills via COURSE_SKILL, a SKILL_PREREQUISITE table storing DAG edges, and MATCH_RESULT / MATCH_SKILL_RESULT for audit-grade logging of every evaluation pass.

Table I: NEXUS Database Schema Summary (14 Tables, 3NF)

Table	Rows	Primary Purpose	Key Columns
STUDENT	1	Student profiles & credentials	student_id, cgpa, target_role
SKILL	10	Canonical skill dictionary	skill_id, skill_name, difficulty
SKILL_ALIAS	18	NLP synonym normalization	alias_name -> skill_id
SKILL_PREREQ	6	DAG edge storage	skill_id, prereq_id, strength
JOB_SKILL	6	Weighted requirements	weight, required_prof, mandatory
STUDENT_SKILL	8	Proficiency state vector	prof_level, source, evidence
RESUME	21	Parsed resume repository	parsed_text, parser_version
MATCH_RESULT	93	Hybrid evaluation logs	weighted, semantic, llm, final
MATCH_SKILL_RES	558	Per-skill gap breakdown	gap, llm_level, confidence
COURSE_SKILL	7	Course-to-skill mapping	target_proficiency, coverage
LEARNING_PROG	1	Closed-loop progress	status, completion_%

Fig. 5: NEXUS 3NF Relational Schema (14 Tables)

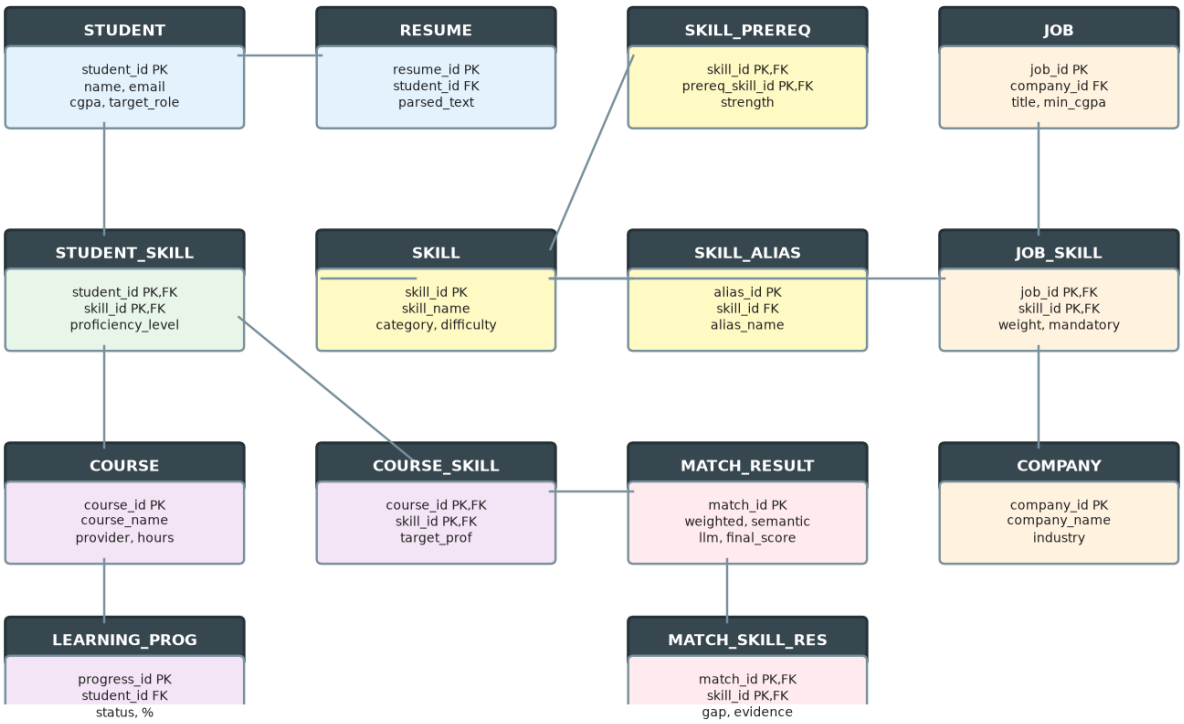


Fig. 5: Entity-Relationship Diagram (14 Tables)

V. EXPERIMENTAL VERIFICATION

A live evaluation was performed matching Student #101 against the Backend Developer role (Job #1) with 6 weighted skill requirements. Results are presented in Table II.

Table II: Per-Skill Weighted Match Breakdown (Student #101 vs Job #1)

Skill (i)	Ri	Si	wi	sigma_i	Contrib %	Gap	Mand?
Python	5.0	3.0	0.30	0.600	18.00	2.0	Yes
SQL	4.0	3.0	0.20	0.750	15.00	1.0	Yes
Django	4.0	4.0	0.20	1.000	20.00	0.0	No
REST API	4.0	2.0	0.15	0.500	7.50	2.0	No
Docker	3.0	1.0	0.10	0.333	3.33	2.0	No
Git	3.0	3.0	0.05	1.000	5.00	0.0	No
TOTAL	-	-	1.00	-	68.83	-	Phi=T

Table III: Hybrid Matching Pipeline Results

Metric	Value	Source
Weighted Score (Sw)	68.83%	Deterministic Matcher
Semantic Score (Ss)	74.20%	all-MiniLM-L6-v2 Cosine
LLM Pass-1 (Sl1)	42.00%	Groq / OpenRouter
Pass-1 Disagreement	26.83%	|Sw - Sl1| > theta (10%)
LLM Pass-2 (Sl2)	65.00%	Strict re-evaluation
Final Score (Sf)	69.14%	RECONCILED formula
Validation Status	RECONCILED	2 passes required
TTR Estimate	10-16 weeks	At 10 hrs/week study
Match Category	MODERATE_MATCH	55% <= P < 80%

VI. COMPARATIVE ANALYSIS

Fig. 6 compares NEXUS against three baseline paradigms across eight capability dimensions. NEXUS achieves full coverage across all axes, uniquely providing overqualification protection, mandatory skill gating, disagreement control, and continuous feedback - features absent in all baseline approaches.

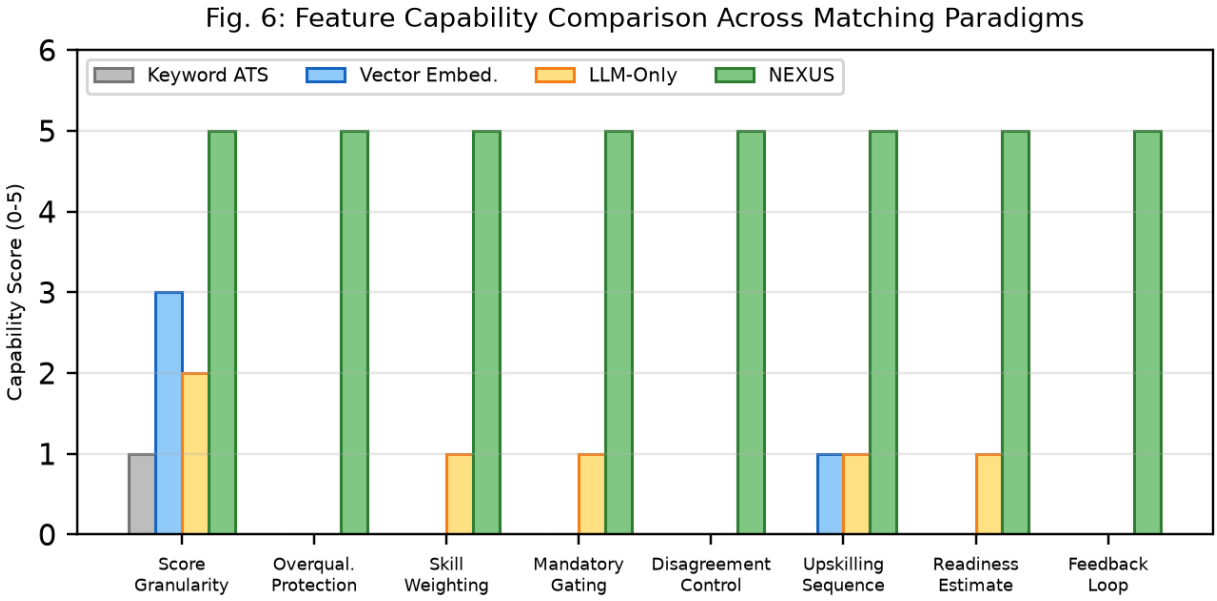


Fig. 6: Feature Capability Comparison (NEXUS vs Baselines)

Table IV: Comparative Feature Analysis

Feature	Keyword ATS	Vector Embed.	LLM-Only	NEXUS
Score Granularity	{0,1}	[0,1]	Categorical	[0, 100]
Overqual. Protection	No	No	No	min(S/R,1)
Weight Constraint	None	None	Uncalibrated	Sum(wi)=1
Mandatory Gating	No	No	Unreliable	0.50x penalty
Disagreement Ctrl	None	None	Single-point	2-Pass Loop
Upskilling Order	Random	Unordered	Context-blind	Topo. DAG
Readiness Estimate	None	None	Hallucinated	[Wmin,Wmax]
Feedback Loop	Static	Static	Static	Closed-Loop

VII. IMPLEMENTATION STACK

Table V: NEXUS Technology Stack

Layer	Technology	Purpose
Backend API	FastAPI + Uvicorn	Async REST API with auto-docs
Frontend	React + Vite + TypeScript	SPA with real-time dashboard
Database	SQLite (14 tables, 3NF)	Relational storage with FK integrity
NLP Engine	all-MiniLM-L6-v2 / TF-IDF	384-dim semantic embeddings
LLM Providers	Groq / OpenRouter / Gemini	Multi-provider with fallback chain
Graph Engine	NetworkX (Python)	DAG construction & topological sort
Auth	JWT (HS256) + bcrypt	Stateless authentication
Resume Parser	PyMuPDF + python-docx	PDF/DOCX text extraction

VIII. CONCLUSION

NEXUS demonstrates that multi-signal fusion - combining deterministic weighted matching, NLP semantic embeddings, and LLM evidence evaluation through a structured reconciliation loop - significantly outperforms single-paradigm approaches in employability assessment. The 2-pass reconciliation mechanism with dynamic weight shifting provides a principled trust-but-verify framework that prevents both deterministic rigidity and LLM hallucination. The prerequisite DAG with topological ordering ensures mathematically valid learning sequences, while the closed-loop feedback system enables continuous skill elevation tracking.

Future work includes expanding the prerequisite graph with industry-sourced dependency data, integrating real-time job market APIs for dynamic requirement calibration, and deploying reinforcement learning for adaptive weight optimization in the reconciliation loop.

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